RAJALAKSHMI ENGINEERING COLLEGE

Harita B 251801095-2025-AIDS

H2

Started on	Tuesday, 25 August 2026, 1:55 PM
State	Finished
Completed on	Saturday, 10 October 2026, 11:16 PM
Time taken	46 days 9 hours
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Convert the following algorithm into a program and find its time

complexity using counter method.

```
void function(int n)
{
    int c= 0;
    for(int i=n/2; i<n; i++)
        for(int j=1; j<n; j = 2 * j)
            for(int k=1; k<n; k = k * 2)
                c++;
}
```

Note: No need of counter increment for declarations and scanf() and count variable printf() statements.

Input:

A positive Integer n

Output:

Print the value of the counter variable

Answer:

```c
#include <stdio.h>

int main()
{
    int n, c = 0;
    int i, j, k;

    scanf("%d", &n);

    i = n / 2;
    c++;   // n / 2

    while (1)
    {
        c++;           // i < n
        if (i >= n)
            break;

        j = 1;

        while (1)
        {
            c++;       // j < n
            if (j >= n)
                break;

            k = 1;

            while (1)
            {
                c++;   // k < n
                if (k >= n)
                    break;

                c++;   // c++
                k = k * 2;
            }

```

```
            j = 2 * j;
        }

        i++;
    }

    printf("%d", c);

    return 0;
}
```

	Input	Expected	Got	
✔	4	30	30	✔
✔	10	212	212	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

Back to Course

RAJALAKSHMI ENGINEERING COLLEGE

Harita B 251801095-2025-AIDS

H2

Started on	Tuesday, 25 August 2026, 1:52 PM
State	Finished
Completed on	Wednesday, 26 August 2026, 11:30 PM
Time taken	1 day 9 hours
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Convert the following algorithm into a program and find its time complexity using the counter method.

```
void function (int n)
{
    int i= 1;
```

```
    int s =1;
```

```
    while(s <= n)
    {
         i++;
         s += i;
     }
}
```

Note: No need of counter increment for declarations and scanf() and count variable printf() statements.

Input:

A positive Integer n

Output:

Print the value of the counter variable

For example:

Input	Result
9	12

Answer: (penalty regime: 0 %)

```c
#include <stdio.h>

int main()
{
    int n;
    int i = 1;
    int s = 1;
    int count = 3;

    scanf("%d", &n);

    while (s <= n)
    {
        i++;
        count++;

        s += i;
        count += 2;
    }

    printf("%d", count);

    return 0;
}
```

	Input	Expected	Got	
✔	9	12	12	✔
✔	4	9	9	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

Back to Course